

# SHUBHARTHAK SANGHARASHA

Applied AI & ML Engineer — Master of Artificial Intelligence & Machine Learning, University of Adelaide  
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## CAREER SUMMARY

Applied AI & ML Engineer with a Master's from the University of Adelaide and hands-on experience spanning robotics, LLMs, and computer vision. Expertise ranges from pretraining language models from scratch and deploying real-world sim-to-real robotic planners (Unitree Go2) to delivering production AI web apps on AWS. Adaptable developer (Python, C++, C, Ruby) driven to build scalable, multi-modal AI systems.

## WORK EXPERIENCE

### Freelance Full-Stack & AI Developer — Self Employed ([devshubh.me](https://devshubh.me))

Aug 2023 – Present

Adelaide, Australia

- End-to-End Client Engineering: architected, built, and deployed production web applications across e-commerce (Aura Boxed Gifts), fintech (Wyndham Financial Group, Auz Finance), and trade services (W13 Projects, BAAZ Electrical Group).
- Applied AI & Voice Assistants: designed and integrated custom multi-modal AI tools, including Aura AI (automated product discovery and checkout) and Wyndham AI (voice-activated mortgage assistance).
- Full-Stack & Growth Systems: built conversion-focused interfaces with dynamic loan calculators, project showcases, lead-generation engines, and performance/SEO optimizations to drive client inquiries.

### Software Engineer Intern — Curve Tomorrow

Jun 2022 – Jul 2023

Parkville, Victoria, Australia

- Built and maintained scalable Ruby on Rails backend features for internal tools and healthcare-focused web applications, including a dynamic CMS for non-technical administrators.
- Developed RESTful APIs, relational database models, CRUD workflows, and secure authentication following Rails MVC architecture and backend best practices.
- Built a user activity logging system and optimized server-side logic and database queries; collaborated with distributed teams in Agile sprints using Ruby, Rails, and PostgreSQL.

## EDUCATION

### Master of Artificial Intelligence & Machine Learning — University of Adelaide

Sep 2024 – Aug 2026

GPA: 6.5 / 7.00 — Courses: Deep Learning Fundamentals, Mining Big Data, ML & AI, Applied NLP, Computer Vision

### B.E., Computer Science & Engineering (AI/ML) — Chandigarh University

Apr 2020 – May 2024

CGPA: 8.39 / 10.00

## PROJECTS

### Action-Conditioned World Models for Autonomous Navigation

Jan 2026 – May 2026

Capstone project: RSSM world model and hybrid CEM planning for autonomous robot navigation (Supervised by Prof. Peng Shi & Dr. Bing Yan)

- Designed and trained a 4.57M-parameter Recurrent State-Space Model (RSSM) on 617K simulated transitions, achieving 0.815 SSIM for multi-step depth prediction.
- Engineered a hybrid Cross-Entropy Method (CEM) planner, reducing collisions by an order of magnitude compared to baseline PPO reactive models.
- Transferred the model to physical Unitree Go2 quadruped hardware, collecting 10.8K real-world transitions and implementing a ChronoDreamer-inspired proximity-gating safety layer for sim-to-real navigation.

### GPT-2 Pretraining & Supervised Fine-Tuning From Scratch

Aug 2025

Large language model pretraining project implementing an end-to-end LLM training pipeline

- Pretrained a 124M-parameter GPT-2 on 5B tokens (FineWeb-Edu), matching OpenAI's baseline with 28.34% HellaSwag accuracy using FlashAttention, FP16 AMP, AdamW, and DDP across multiple GPUs.
- Implemented SFT and DPO alignment pipelines, deploying the fine-tuned LLM via an interactive Gradio app.

### Apsara: Multimodal AI Assistant & RAG Engine

Jan 2025

Open-source, cross-platform AI desktop assistant with voice, vision, and automation, with an Android successor in development

- Developed an open-source cross-platform desktop/mobile AI agent featuring voice, computer vision, and OS-level automation.
- Built a custom Retrieval-Augmented Generation (RAG) system using LangChain to query unstructured PDFs and web data with local (Ollama) and cloud (OpenAI/Gemini) embeddings; developing Apsara Dark (Android successor) with real-time screen awareness and vision-based tap targeting across 5 agentic architectures.

### Face Mask Detection

Multi-face mask detection using TensorFlow & MobileNetV2 transfer learning, with automated email alerts on violations.

### NodeMCU Smart Car

Wirelessly controlled IoT car built on a NodeMCU ESP8266 with Blynk for remote Android/web control.

## SKILLS

Python, C++, C, Java, Ruby, PyTorch, OpenCV, TensorFlow, NLP, LangChain, Git, LLMs, AWS, Reinforcement Learning (RSSM/CEM), World Models, TTS, Speech Recognition, React, NumPy, Selenium, Unix, Audio Signal Processing, Pandas, Ruby on Rails, MySQL

## CERTIFICATIONS

Machine Learning • Supervised Machine Learning: Regression and Classification • Advanced Learning Algorithms • Tools for Data Science • Databases and SQL for Data Science with Python • Linux Fundamentals